LAGOS BUSINESS SCHOOL · MEMBA 14 · EEGB · PART IV

SESSION 15

Corporate Macro Strategy: Hedging & Resilience



From markets, to macro, to decisions that matter — in Nigeria and the world.

Reading: Acharya, Almeida, Amihud & Liu — Corporate Resiliency and the Choice Between Financial and Operational Hedging (NBER) · Facilitator: Sumendra Dash

RECAP

Session 14 in Three Ideas



Country risk

Score markets on political, policy, external and banking risk — then compare like with like.



Scenarios

Build three to four plausible futures, each with signposts you can watch.



Stress tests

Ask what the plan looks like in the bad scenario — and what you would do differently.

TODAY'S PLAN

Session 15 at a Glance — 75 Minutes

From knowing your macro risks to managing them — with contracts, operations and capital.



Recap & Hook

Why a profitable miller nearly lost its margin

Measuring Exposure

Which macro numbers really drive the P&L?

Hedging Choices

Natural vs financial hedges, forwards and real options

Lab: Hedging Plan

Heatmap, hedge ratio and dashboard

BY THE END OF TODAY

Session 15 Learning Objectives

You will be able to:

01

Measure Macro Exposure

Trace how the exchange rate, commodity prices and interest rates flow into revenue, costs and margin.

02

Choose the Right Hedge

Compare natural, operational and financial hedges, and pick a sensible hedge ratio — not automatically 100%.

03

Build Resilience

Use a risk heatmap, real options and leading indicators to prepare before the shock arrives.

THE HOOK

Profitable — Until the Naira Moved

“We buy wheat in dollars and sell bread flour in naira. In February 2024 our gross margin was zero.”

The miller's managers did nothing wrong in the factory. Three macro questions decided the year:

How exposed are we?

How much of our cost is set in dollars, and how fast can our prices follow?

What can we lock in?

Can we fix the naira cost of next quarter's wheat today — and at what price?

What else can we change?

Sourcing, pricing, stock, debt — operations can hedge too.

The Flour Miller's Hedging Plan

The situation

A Lagos flour miller imports all its wheat, paid in US dollars about **three months** after ordering. It sells flour to bakers and noodle makers in naira, and can raise prices only gradually.

The decision question

Which macro risks matter most, **how much of its dollar cost should it hedge**, and what else should it do to protect its margin?

The data we have

miller_exposure_monthly.csv — 84 months (2019–2025): wheat price, exchange rate, volumes, costs, flour price, revenue and margin. Plus ng_macro_monthly.csv for rates and FX. *Stylised teaching data.*

What a good answer looks like

A one-page **hedging plan**: a risk heatmap, a recommended hedge ratio with its cost, the operational hedges, and five **leading indicators** to watch.

PART

01

Measuring Exposure

You cannot manage a risk you have not measured.

How Macro Risk Reaches the P&L

Transaction exposure

A dollar bill you must pay, or receive, on a known date. Next quarter's wheat shipment is the miller's biggest one.

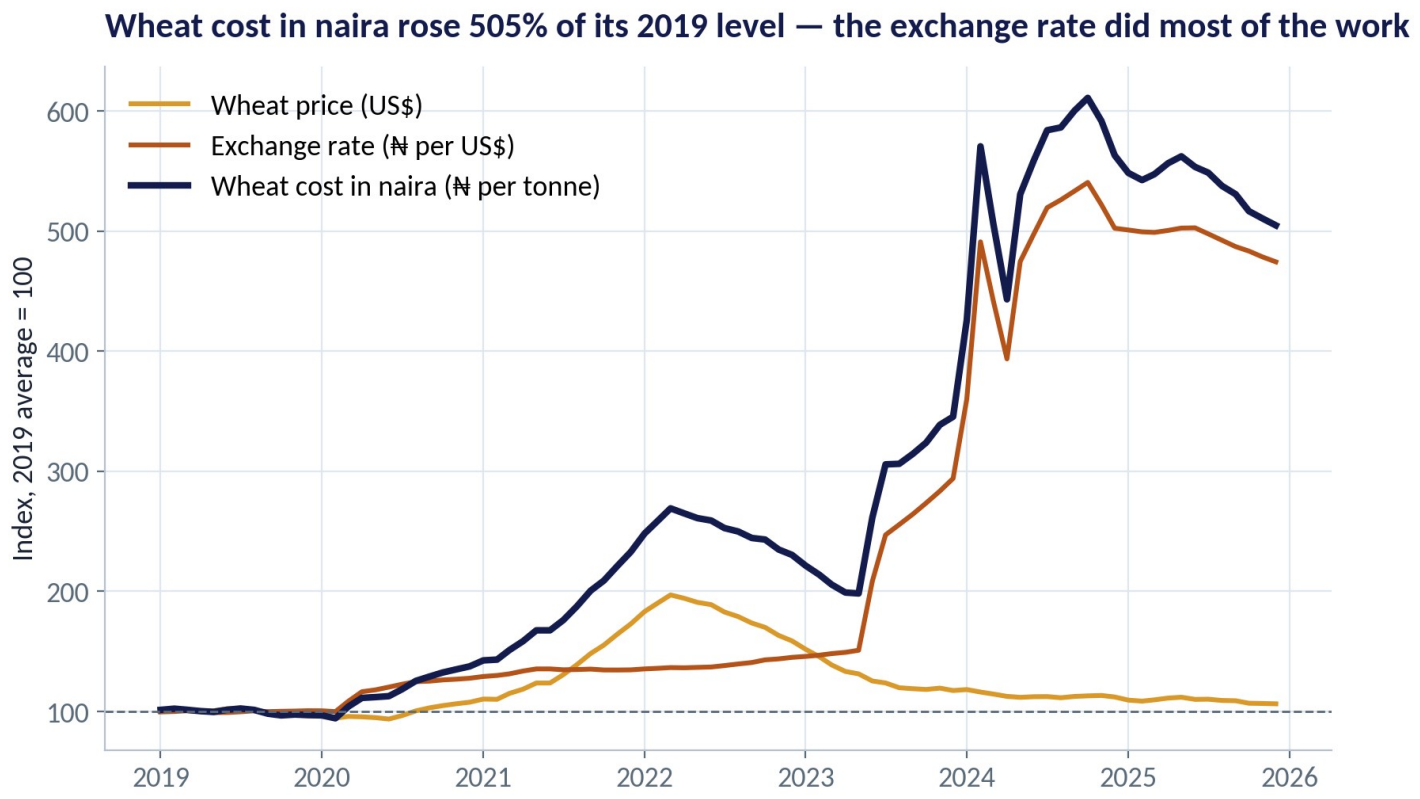
Translation exposure

Dollar assets or debts on the balance sheet revalued in naira. A dollar loan grows overnight after a devaluation.

Economic exposure

How the macro environment changes your competitiveness and demand over years — even if you never touch a dollar.

Wheat in Dollars, Dollars in Naira



Source: EEGB stylised teaching data (data/). Not official statistics.

Fig S15-F01 · replicate: s15_figures.R / .py

WHAT TO SEE

- By December 2025 the naira cost of a tonne of wheat was about **five times** its 2019 level (index 505).
- The dollar price of wheat ended only **6% above** 2019, after a spike in 2022.
- The exchange rate index reached **474**: the naira did most of the damage.
- Lesson: the miller's real exposure is to the **naira**, not to wheat.

How Much Came From Each Source?

IN PLAIN ENGLISH

A naira cost of an imported input is **dollar price × exchange rate**. In logs, the two parts simply add up, so you can split any rise into a 'commodity' part and a 'currency' part.

EXAMPLE

2019 → 2025 averages: wheat **+9** log-points, naira **+160** log-points. Total rise **438%**, of which the exchange rate explains **95%**.

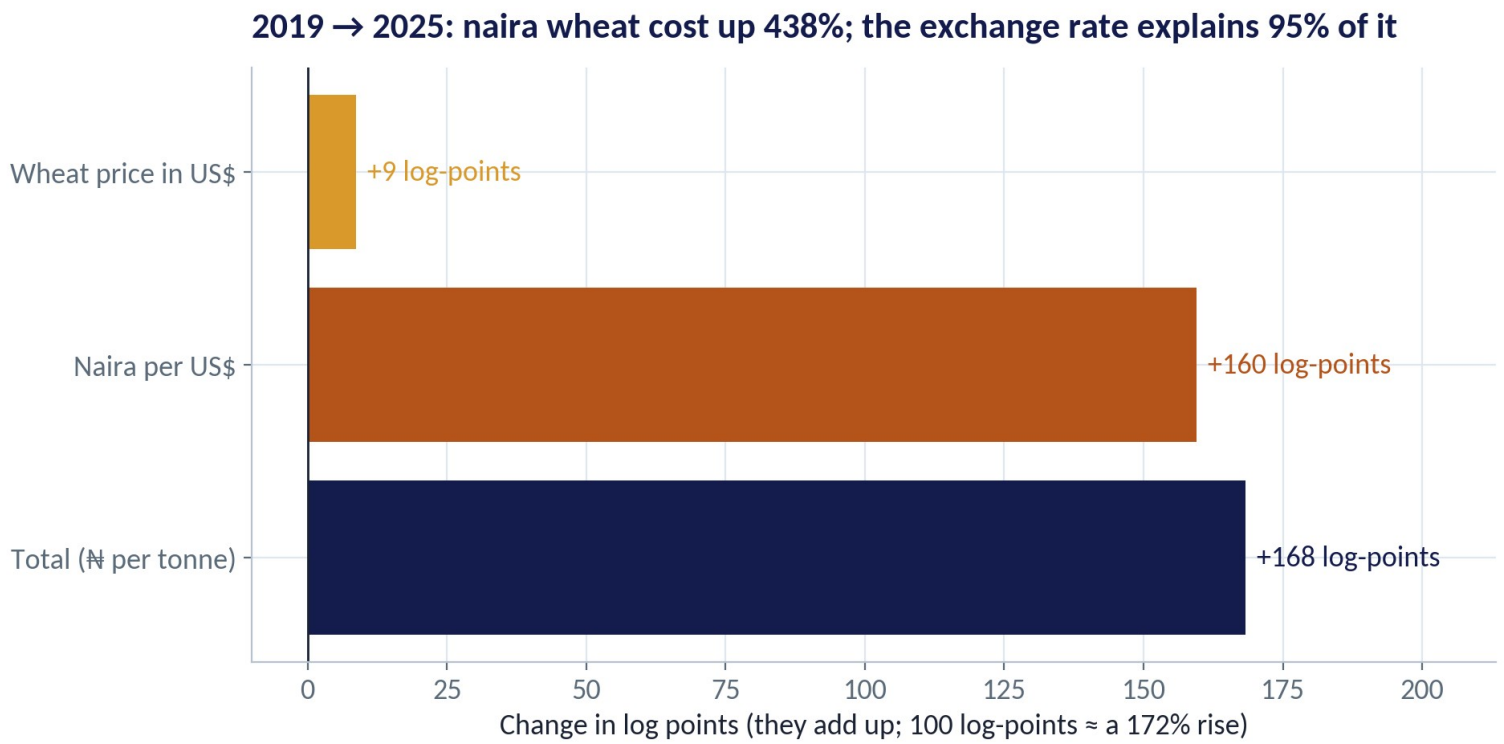
THE FORMULA

$$\Delta \ln(₦ \text{ cost}) = \Delta \ln(\text{US\$ price}) + \Delta \ln(₦ \text{ per US\$})$$

WHY A MANAGER CARES

It tells you which hedge matters. Buying wheat futures in Chicago would have done little; managing the naira exposure was everything.

The Exchange Rate Did the Damage



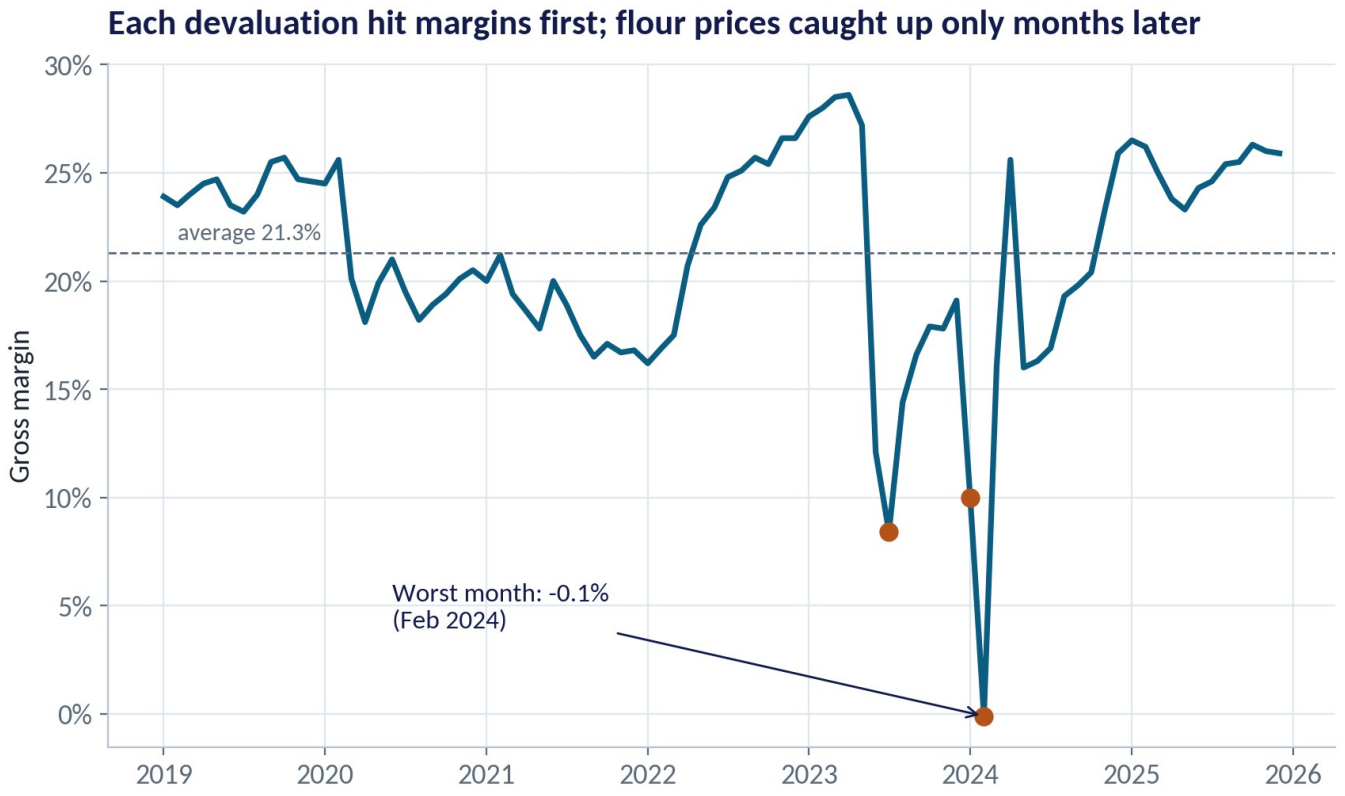
Source: EEGB stylised teaching data (data/). Not official statistics.

Fig S15-F02 · replicate: s15_figures.R / .py

WHAT TO SEE

- Wheat price in US$: **+9** log-points.
- Naira per US$: **+160** log-points.
- Together: naira wheat cost up **438%** between the 2019 and 2025 averages.
- The currency explains **95%** of the increase.

Costs Move First, Prices Follow



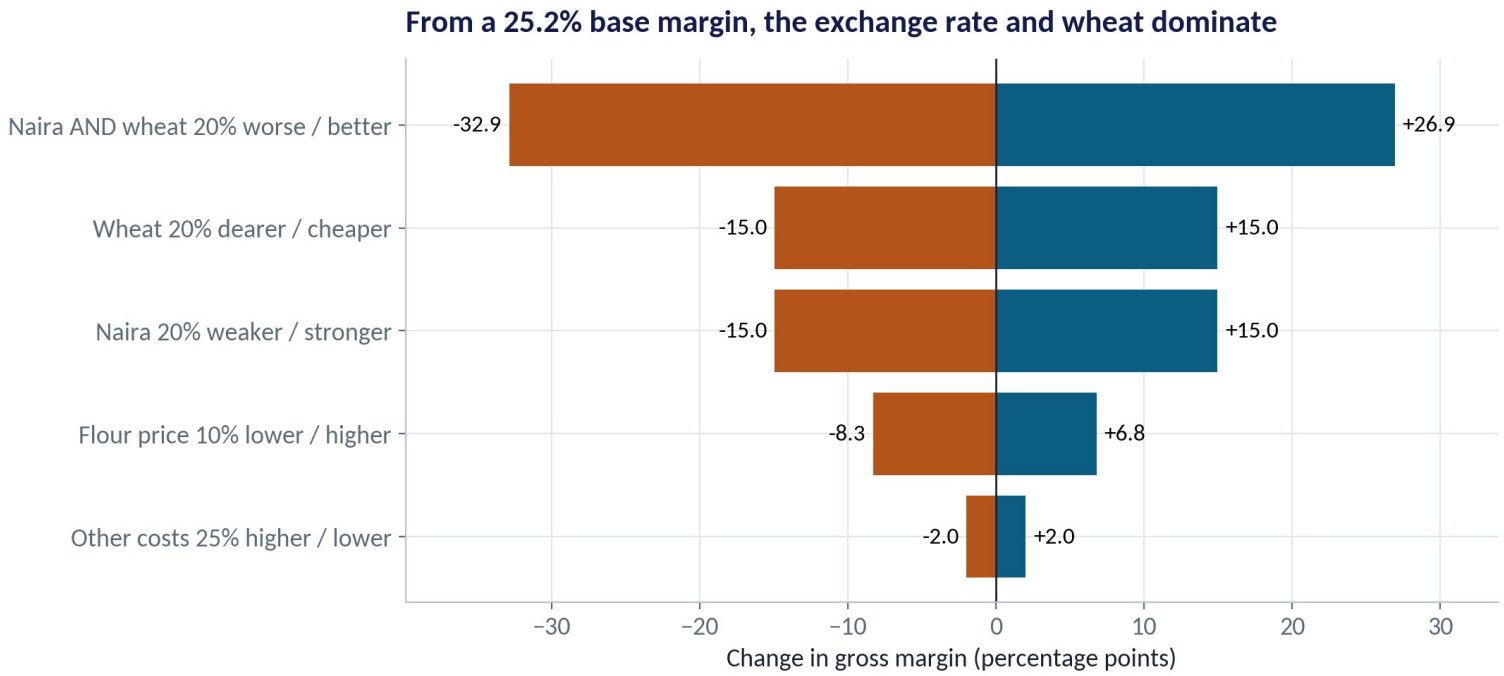
Source: EEGB stylised teaching data (data/). Not official statistics.

Fig S15-F03 · replicate: s15_figures.R / .py

WHAT TO SEE

- Average gross margin 2019–2025: **21.3%**.
- Each devaluation hit the margin **before** flour prices could follow.
- Worst months: **−0.1%** in February 2024, **8.4%** in July 2023 and **10.0%** in January 2024.
- Margins recovered as prices caught up — a **natural hedge**, but a slow one.

A Sensitivity 'Tornado'



Source: EEGB stylised teaching data (data/). Not official statistics.

Fig S15-F06 · replicate: s15_figures.R / .py

WHAT TO SEE

- Starting point: 2025 average margin of **25.2%**.
- A 20% weaker naira, or 20% dearer wheat, each cuts the margin by **15 points**.
- Both together: **−32.9 points** — the miller would make a loss.
- Other costs matter far less (**±2 points**). Spend management time where the tornado is widest.

Quick Check: What Would You Say?

?

The miller's naira wheat cost rose 438% between 2019 and 2025. Wheat in dollars rose about 9%. Which hedge would have helped most?

A. Buying wheat futures to fix the dollar price

B. Fixing the naira cost of dollars in advance (FX forwards) and passing costs on faster

C. Borrowing more in dollars at low rates

D. Holding larger stocks of flour

Discuss with your neighbour for 60 seconds — answer and reasoning in the speaker notes.

PART

02

Hedging Choices

Contracts, operations and capital.

Natural Hedges vs Financial Hedges

Natural / operational hedges

Change the business so risks offset

- Price flour off the dollar cost (faster pass-through)
- Earn some revenue in dollars (export to ECOWAS)
- Source local inputs (cassava, sorghum blends)
- Hold more stock when the naira looks weak
- Borrow in the currency you earn

Financial hedges

Sign a contract that pays off when the shock hits

- FX forwards: fix the naira cost of future dollars
- Futures and swaps (where markets exist)
- Commodity futures for wheat
- Options: insurance, for a premium
- Cost, availability and tenor vary — ask your bank

*Research on corporate resiliency finds firms often prefer **operational** hedges because financial hedges need cash or credit lines just when they are scarce.*

What a Naira Forward Should Cost

IN PLAIN ENGLISH

A forward fixes today the rate at which you will buy dollars later. Banks price it so nobody can earn a free profit: the naira forward is **weaker** than spot by roughly the **interest-rate gap** between naira and dollars.

EXAMPLE

December 2025: spot **₦1,454**, MPR **27%**, US rate about **3.75%** → 3-month forward ≈ **₦1,539**, a premium of **5.8%** a quarter.

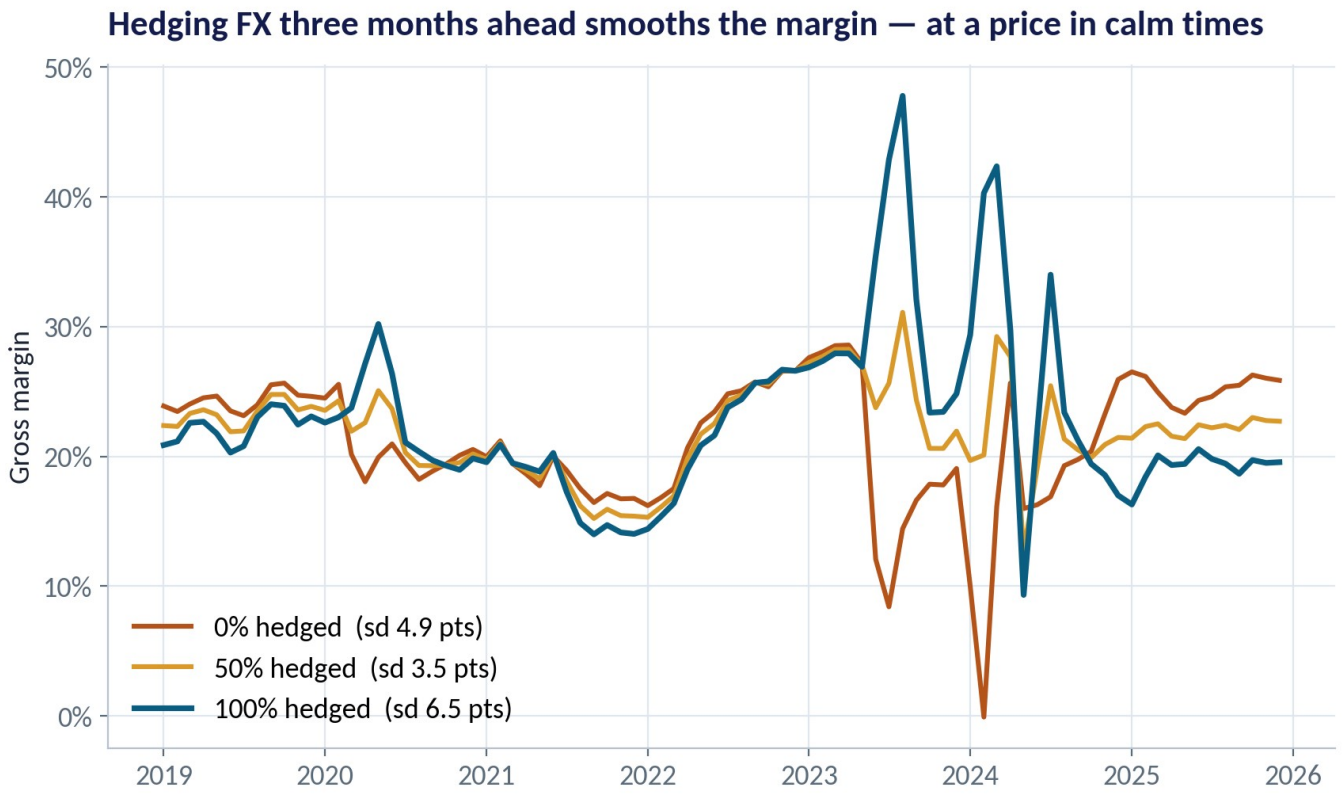
THE FORMULA

$$\text{Forward} \approx \text{Spot} \times (1 + (i₦ - i\$) \times \text{months} \div 12)$$

WHY A MANAGER CARES

Hedging is not free: at a 27% naira rate, rolling three-month cover costs over 20% a year in calm times. It pays off only when the naira falls by more than the premium.

Hedging 0%, 50% or 100% of Next Quarter's Dollars



Source: EEGB stylised teaching data. Forward = spot × (1 + (MPR − US policy rate) × 3/12). Illustrative only.

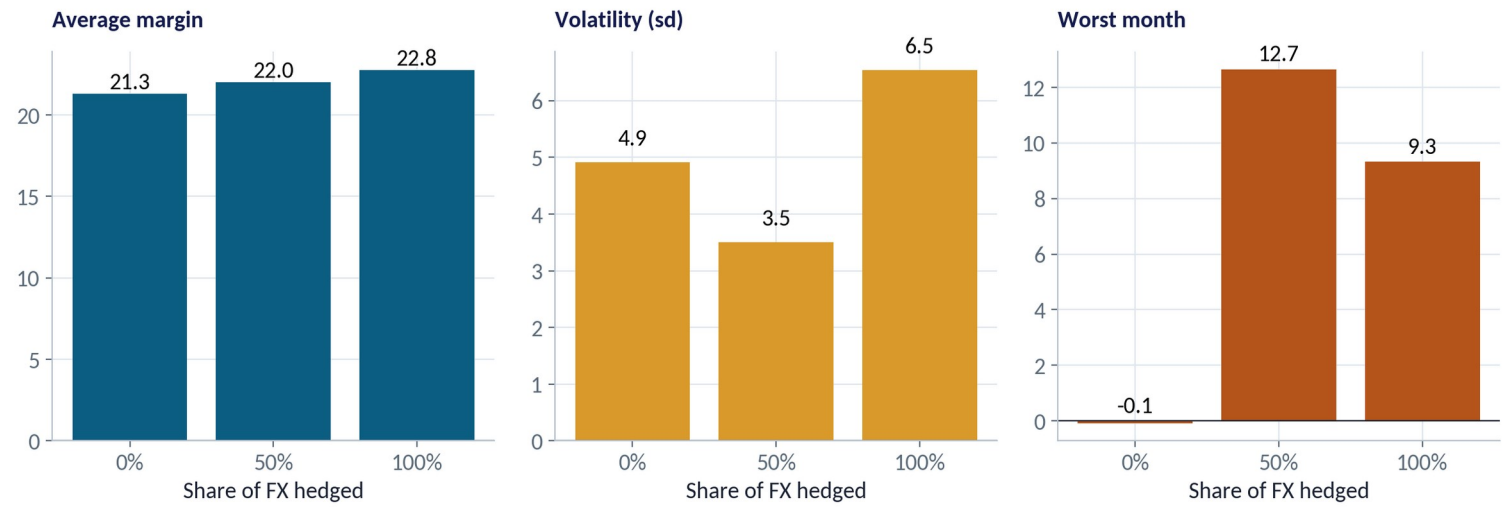
Fig S15-F04 · replicate: s15_figures.R / .py

WHAT TO SEE

- Each month the miller can lock in the rate for wheat it will pay for in **three months**.
- Unhedged, the margin fell to **−0.1%** in the 2024 devaluation.
- Fully hedged, the margin **jumped** in the same months — prices rose while costs were locked.
- In calm years hedging cost margin: average **21.8%** unhedged vs **20.4%** fully hedged.

Why 100% Is Not the Safest Hedge

A 50% hedge gave the steadiest margin; 100% over-hedged because flour prices already pass on costs



Source: EEGB stylised teaching data (data/). Not official statistics.

Fig S15-F05 · replicate: s15_figures.R / .py

WHAT TO SEE

- **50% hedged** had the lowest volatility: **3.5 points** vs 4.9 unhedged and 6.5 fully hedged.
- Its worst month was **12.7%**, against −0.1% unhedged.
- 100% **over-hedged**: flour prices already follow the naira, so a full hedge creates a new mismatch.
- The right hedge ratio covers the exposure that your **prices cannot pass on**.

Flexibility Is a Hedge Too

IN PLAIN ENGLISH

A **real option** is the right, but not the obligation, to change course later — like having a second supplier ready, or a plant that can switch inputs. It costs something today and pays off when the world turns against you.

EXAMPLE

Testing a **cassava-wheat blend** costs ₦ today. If the naira falls 20%, switching 10% of wheat to local cassava cuts the dollar bill exactly when it hurts most.

THE FORMULA

Value of flexibility = value with the option − value without it

WHY A MANAGER CARES

Options to delay, expand, switch or exit are often cheaper and longer-lasting than financial contracts, which may not exist for long tenors in naira.

Matching the Hedge to the Risk

Risk	Financial hedge	Operational hedge	Capital buffer
Naira devaluation	FX forwards / futures for next 3–6 months	Faster price pass-through; dollar export sales	Cash buffer; avoid unhedged US$ debt
Wheat price spike	Wheat futures or fixed-price contracts	Blends with local grains; supplier diversity	Working-capital line
Interest-rate rise	Fixed-rate loans or rate swaps	Shorter credit to customers	Lower leverage
Port & shipping delays	Freight contracts, insurance	Higher safety stock; second port	Inventory finance
Demand slump	—	Smaller packs; product mix; cost flexibility	Liquidity to ride it out

*Most firms use all three columns. Financial hedges buy **time**; operational hedges change the **structure** of the exposure.*

Quick Check: What Would You Say?

?

The miller's prices follow the naira with a lag of a few months. The board asks to hedge 100% of FX for the next year. What is the best response?

- **A** Agree — full hedging removes all risk
- **B** Hedge only the part of the exposure that prices cannot pass on, over the lag period — roughly half for a few months — and review quarterly
- **C** Refuse all hedging — the naira forward premium is too high
- **D** Hedge 100% but for five years to save on fees

Discuss with your neighbour for 60 seconds — answer and reasoning in the speaker notes.

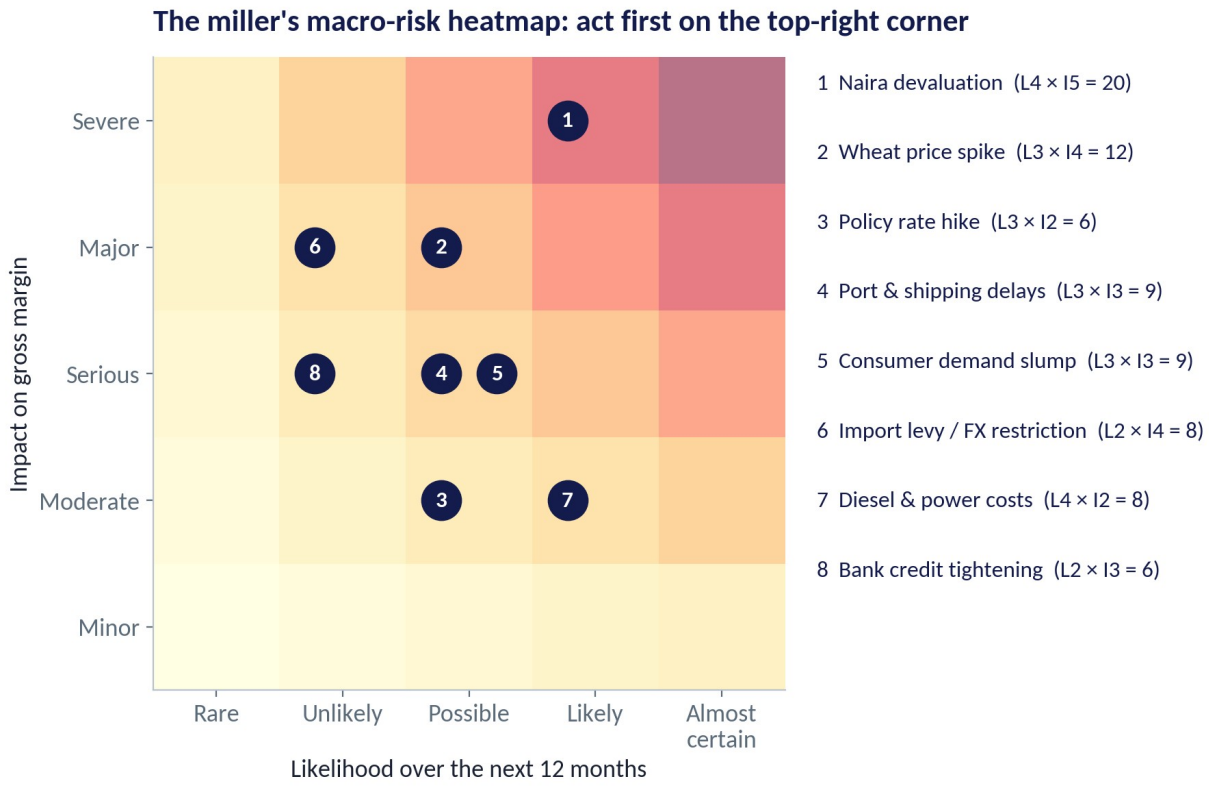
PART

03

Building Resilience

Prepare before the shock arrives.

The Macro-Risk Heatmap



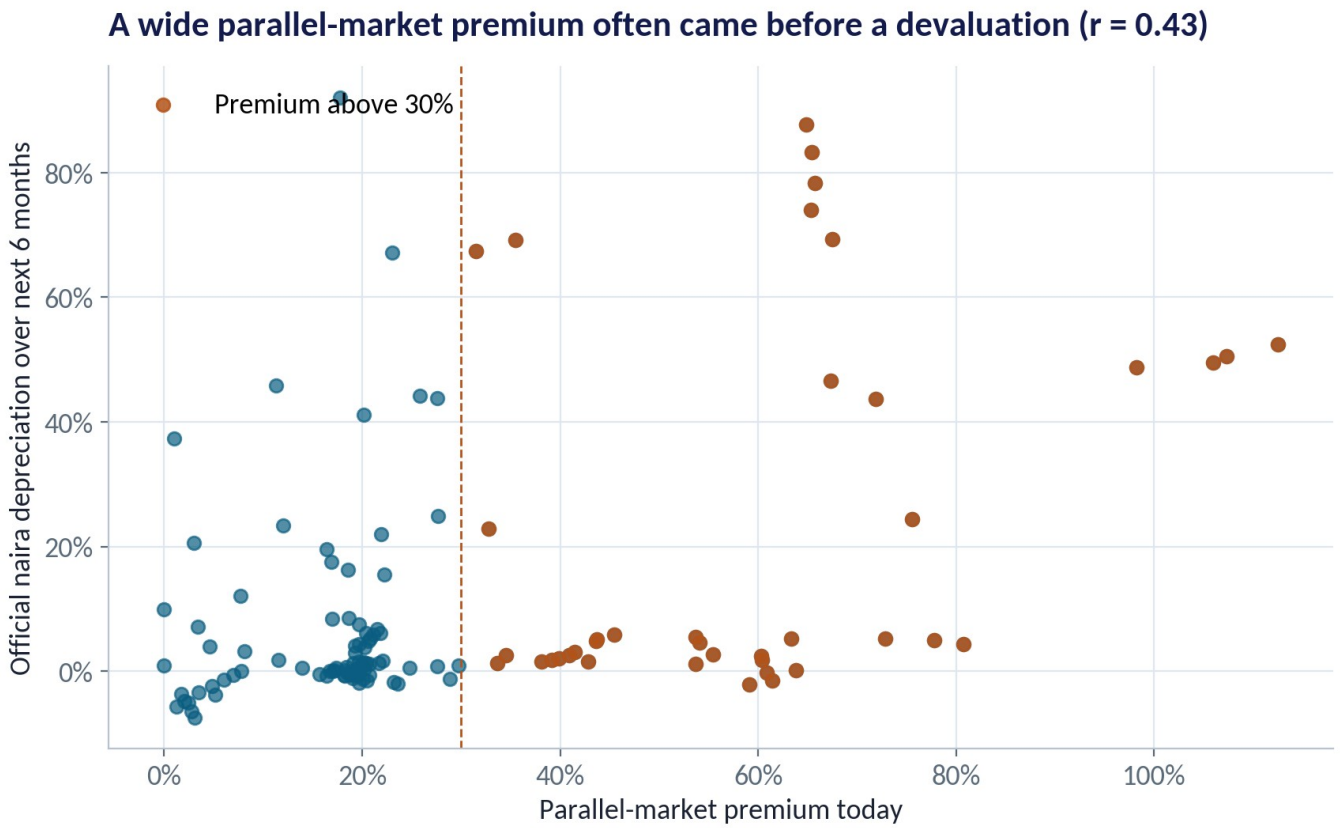
Scores are management judgements, informed by the sensitivity analysis (F06). Illustrative.

Fig S15-F07 · replicate: s15_figures.R / .py

WHAT TO SEE

- Score each risk for **likelihood** and **impact** (1–5); the product ranks them.
- **Naira devaluation** (4 × 5 = 20) and **wheat spikes** (3 × 4 = 12) lead.
- Impact scores come from the sensitivity tornado, not gut feel.
- Review the map every quarter — scores change as signposts move.

Signals That Warn Before the Shock



Source: EEGB stylised teaching data (data/). Not official statistics.

Fig S15-F08 · replicate: s15_figures.R / .py

WHAT TO SEE

- When the parallel premium was **above 30%**, the naira fell on average **23.4%** over the next six months.
- When it was below 30%, the average was **7.0%**.
- The correlation is moderate (**r = 0.43**): a warning light, not a forecast.
- Raise the hedge ratio, or stock up, when several warning lights turn on together.

Five Leading Indicators for the Miller

FX premium

Gap between parallel and official rates — a sign of dollar rationing.

Reserves

CBN reserves and their trend: the central bank's ability to defend the naira.

Oil price

Brent drives Nigeria's dollar earnings and reserves.

Rate gap

MPR minus US rates: sets the cost of a forward hedge.

Wheat futures

Global wheat prices three to six months ahead.

How Companies Get Hedging Wrong

THE TRAP	THE BETTER HABIT
✗ Hedging 100% of everything, forever.	✓ Hedge the **residual exposure** your prices cannot absorb, over the **lag** period.
✗ Judging a hedge by whether it 'made money'.	✓ Judge it by whether it **reduced volatility** and protected the plan.
✗ Ignoring the forward premium at high naira rates.	✓ Know the **cost of cover** and when it is worth paying.
✗ Relying only on contracts that may not be available.	✓ Build **operational hedges**: pricing, sourcing, stock, currency of debt.
✗ Waiting for the devaluation before acting.	✓ Watch **leading indicators** and set trigger levels in advance.

HANDS-ON LAB

Lab: A Hedging Plan for the Miller

YOU WILL NEED

Data

`data/miller_exposure_monthly.csv`
`data/ng_macro_monthly.csv`
`data/global_policy_rates.csv`

R script

`sessions/s15/R/s15_lab.R`

Python script

`sessions/s15/python/s15_lab.py`

15 minutes

1. Map the exposure: which costs are in dollars, and how did the **margin** behave?
2. **Decompose** the rise in naira wheat cost into wheat and exchange-rate parts.
3. Build a **3-month forward price** from the interest-rate gap.
4. Simulate hedge ratios of **0, 25, 50, 75 and 100%** and compare average, volatility and worst month.
5. Run the **sensitivity** test and score your own company's top five risks.

Deliverable: a one-page hedging plan: heatmap, hedge ratio with its cost, two operational hedges, five indicators.

WORDS TO OWN

Today's Vocabulary in Plain English

Transaction exposure	A known foreign-currency payment or receipt on a future date.
Translation exposure	Revaluation of foreign-currency assets and debts on the balance sheet.
Economic exposure	Long-run effect of macro changes on competitiveness and demand.
Natural hedge	An offsetting exposure inside the business, e.g. dollar revenue against dollar costs.
Forward contract	An agreement today on the rate for a future currency exchange.
Forward premium	How much weaker the forward rate is than spot; set by the rate gap.
Hedge ratio	Share of an exposure covered by a hedge.
Real option	The right to change course later: delay, expand, switch or exit.
Sensitivity analysis	How much an outcome moves when one input changes.
Leading indicator	A signal that tends to move before the outcome you care about.

KEY TAKEAWAYS

Session 15 in 5 Ideas

01 **Measure** exposure first: decompose costs into commodity and currency parts.

02 For Nigerian importers, the **exchange rate** usually dominates.

03 Hedge the exposure your **prices cannot pass on** — 100% can over-hedge.

04 Forward cover costs roughly the **interest-rate gap**; know when it is worth paying.

05 Build **operational resilience** and watch **leading indicators** with pre-agreed triggers.



NEXT › Session 16: Capstone & Review

Before next session: Prepare for the capstone: choose your industry, draft the one-page macro memo to your board using the template on the course site, and bring three charts as evidence.